

NuMicro[®] Family VSCode Quick Start Guide

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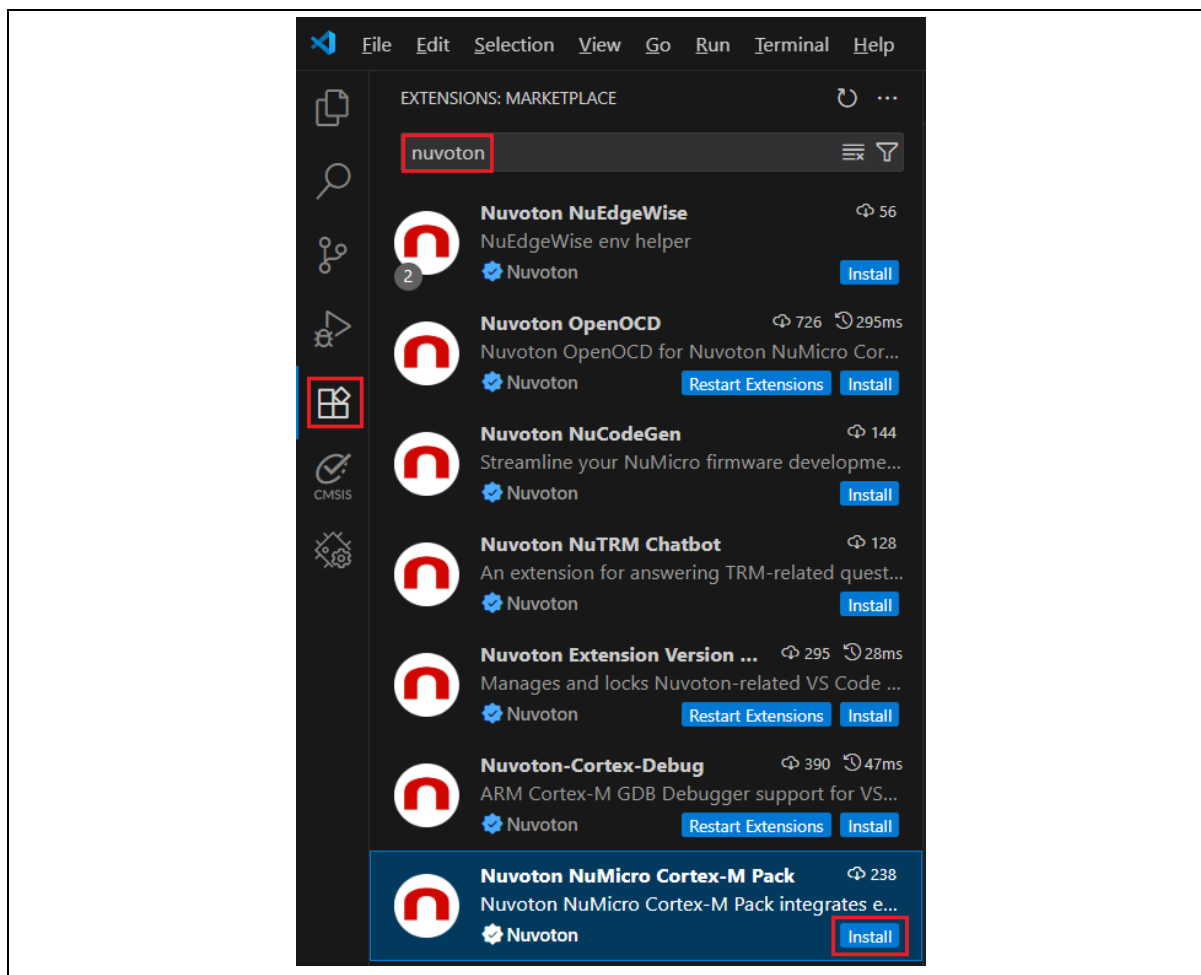
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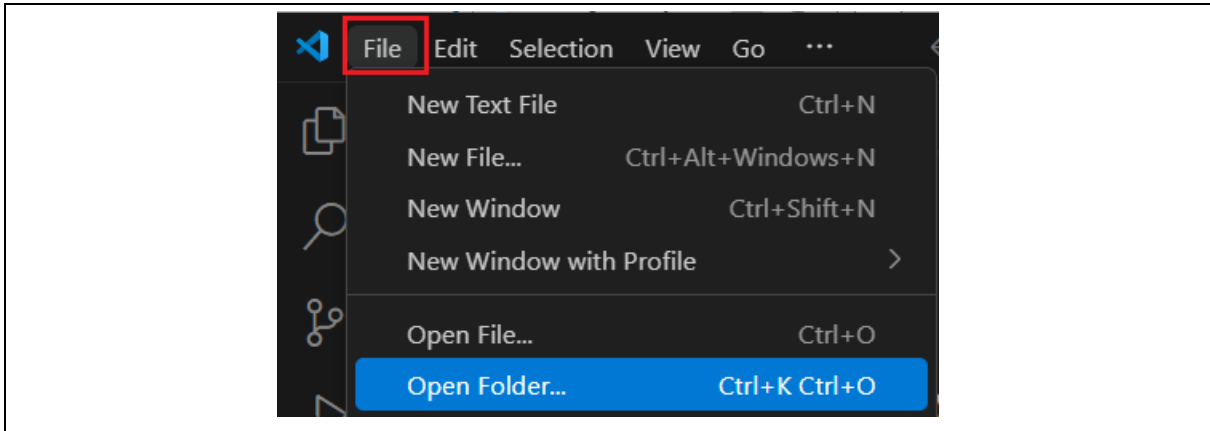
1 INSTALLATION VSCODE AND EXTENSIONS

1. Download VSCode from <https://code.visualstudio.com> and install it.
2. Launch VSCode and click Extensions in the Activity Bar.
3. Text “**Nuvoton NuMicro Cortex-M Pack Extension**” in search bar. Click the “**Install**” button to install.

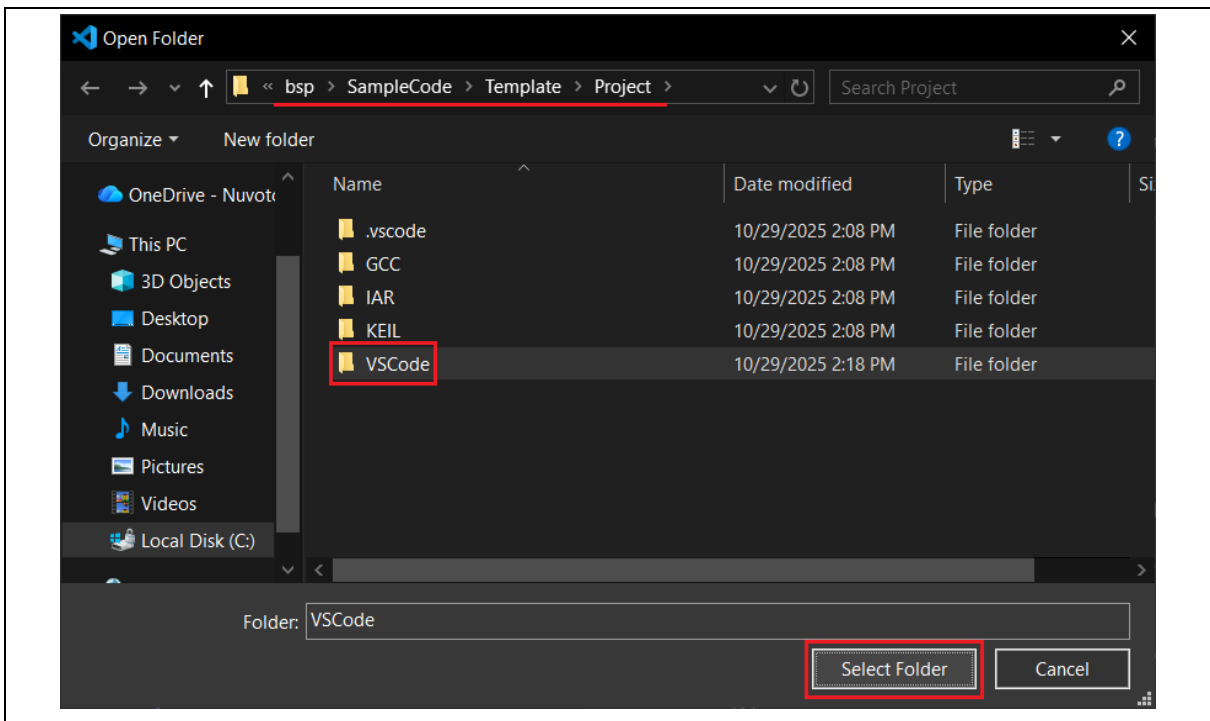


2 GET STARTED WITH A SAMPLE PROJECT

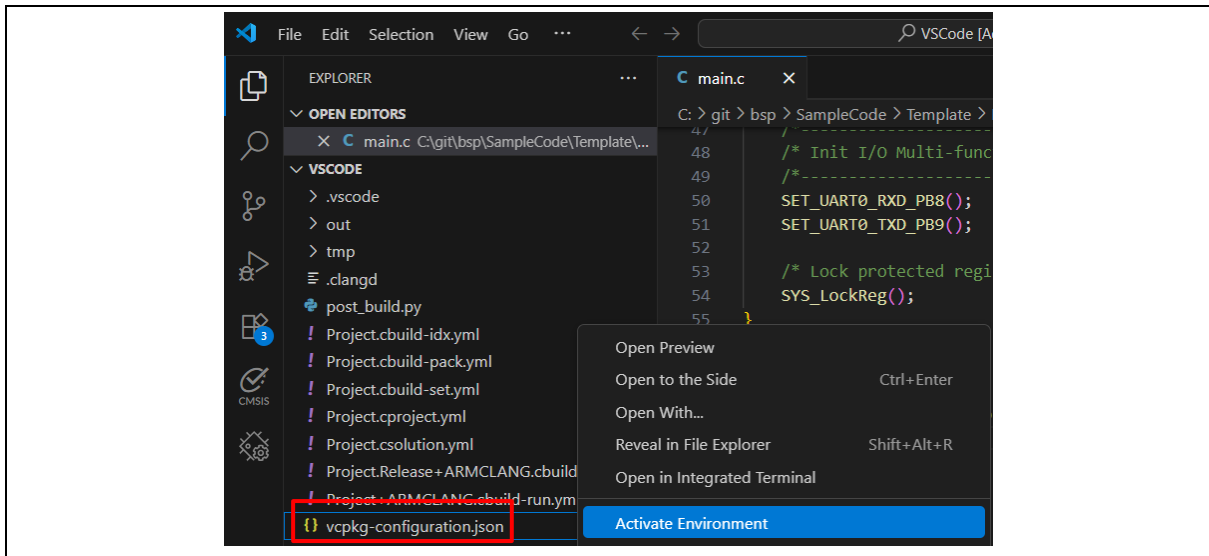
1. Click "File" menu and select "Open Folder" in the toolbar.



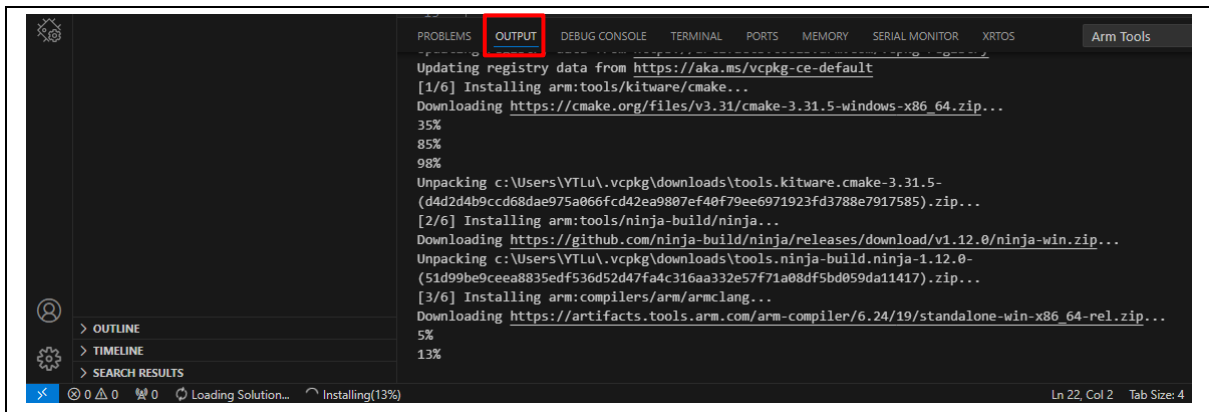
2. Select the "VSCode" folder of a sample project to open the project.



3. Right-click “**vcpkg-configuration.json**” file and select “**Activate Environment**” command. (Please use private network connection.)

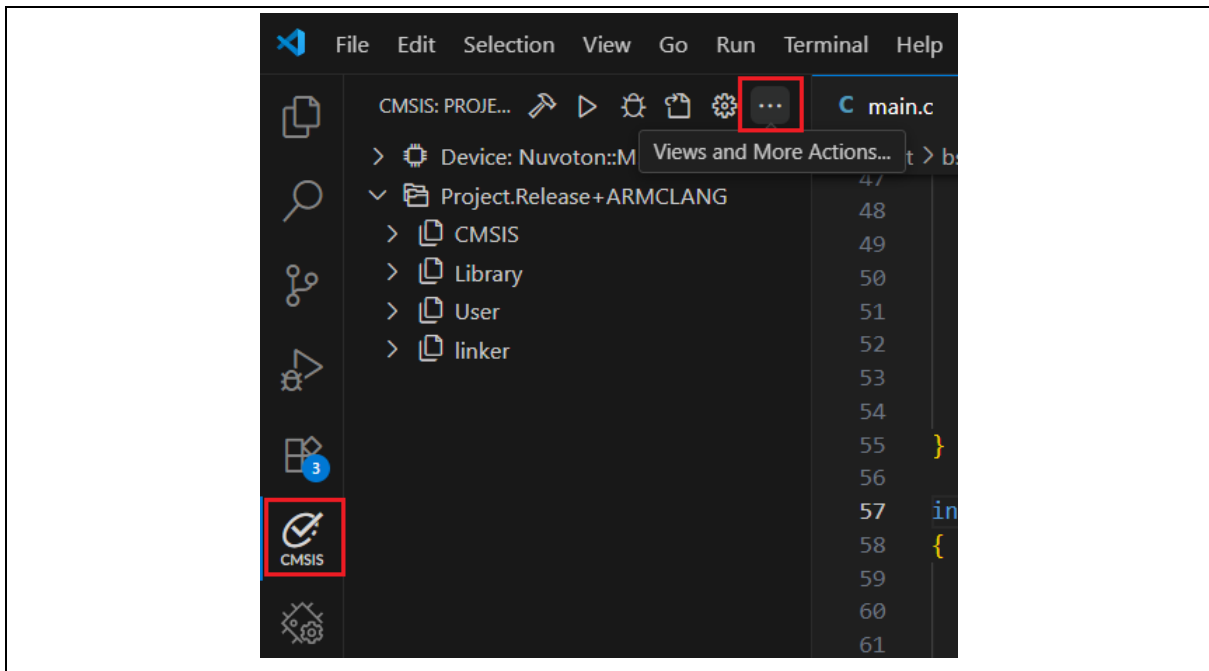


4. Check **OUTPUT** terminal at the bottom. It will download and install the requires tools.

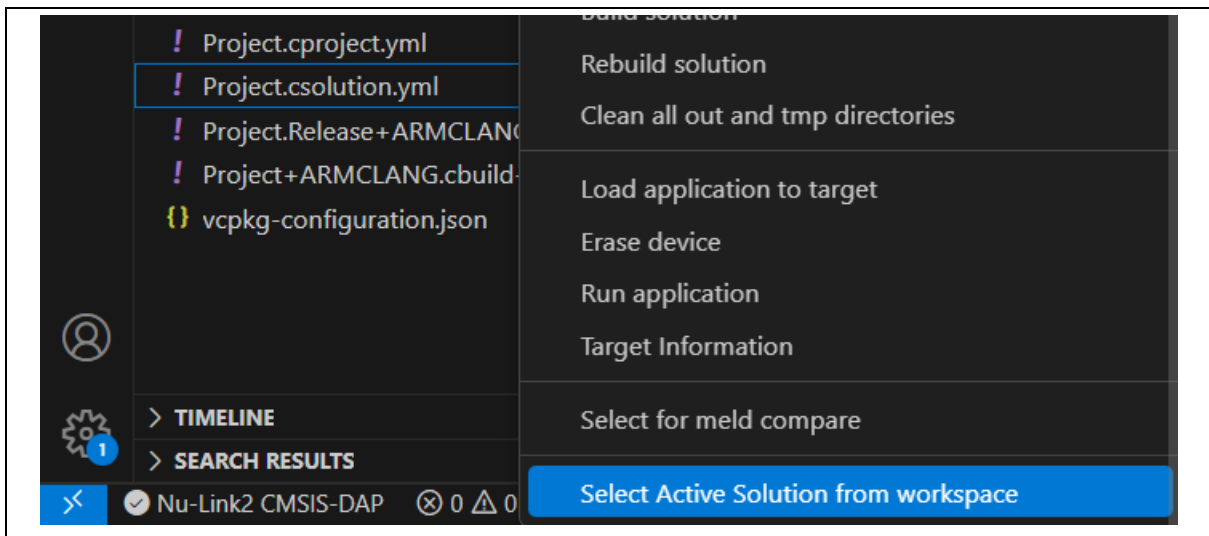


5. Click **CMSIS** icon in the Activity Bar.

Then click **“Views and More Commands”** button.

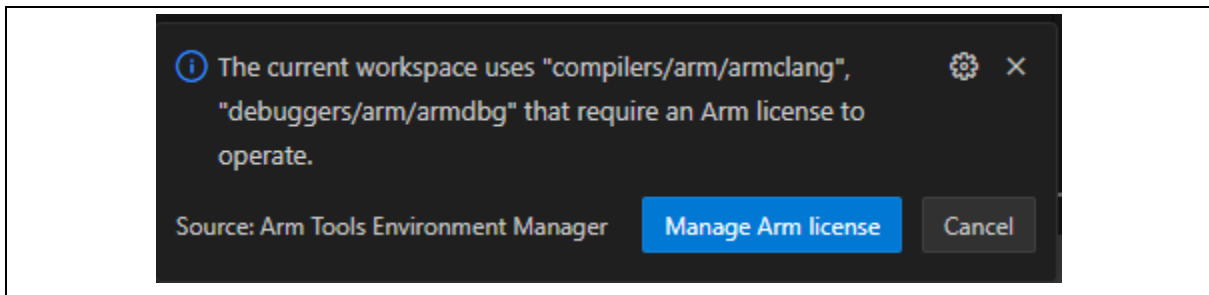


6. Select **“Select Active solution from workspace”** command.

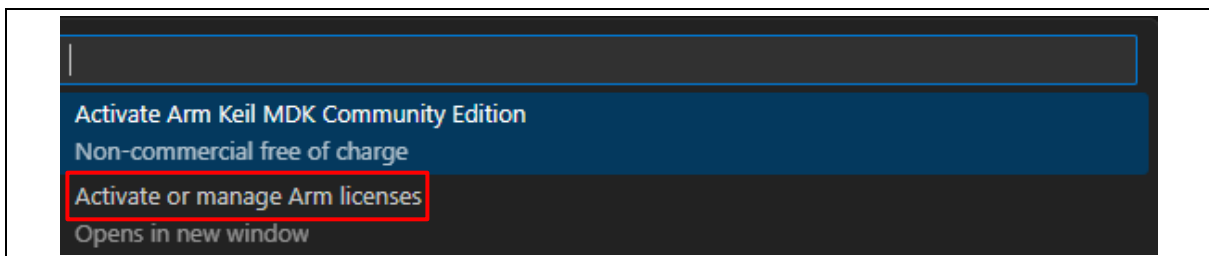


3 MANAGE ARM LICENSE

1. Click **"Manage Arm License"** button in the pop-up window at the bottom right.

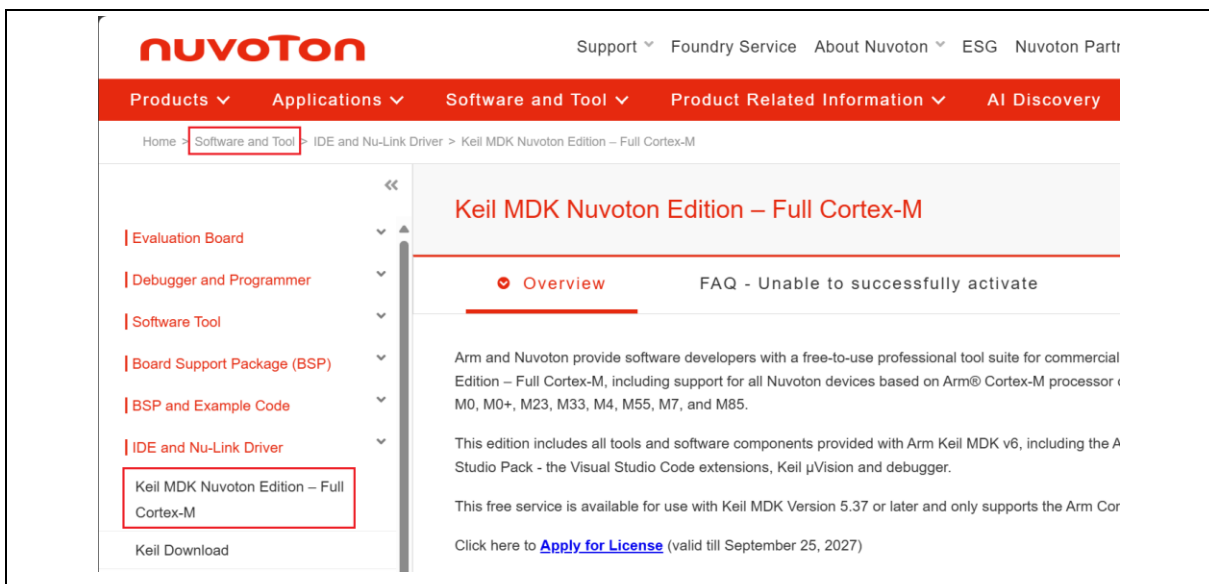


2. In the search bar at the top, select **"Activate or manage Arm licenses"** command from the available options.



3. Get Keil MDK License ID code.

- 1) Navigation to [Official Website](#).



2) Fill out the form.

Apply for Keil MDK NuvoTon Edition – Full Cortex-M

First Name*

Last Name*

Company / Organization*

Job Title*

Region / Country or region*
 -- Select Region -- -- Country or region --

State / Province

Email*

Phone*

Industry
 -- Industry --

Application*
 -- Application --

Series*
 -- Select Line -- -- Select Category --

Part No.

NUVOTON Privacy Policy

NUVOTON Privacy Policy
 Last modified: June 01, 2023

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By clicking "I agree" and I accept Nuvoton's Privacy Policy

I agree to be my Nuvoton member

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Submit

3) Check mailbox and fill in the License Server.

Arm License Management Utility

Activate your product

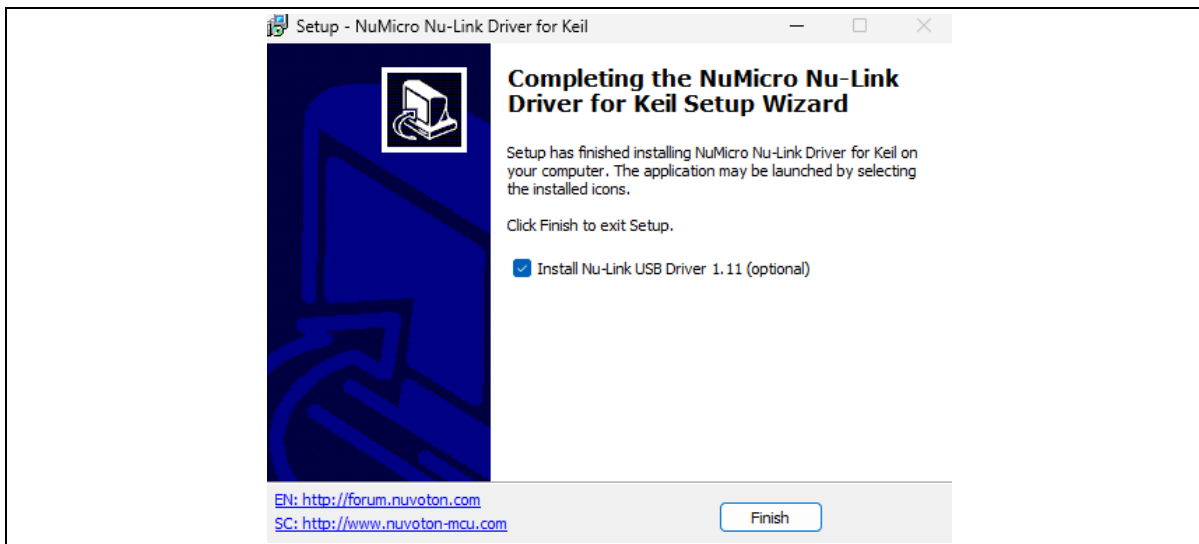
Activate with: ☐ Activation Code ☒ License Server

Enter your license server address:

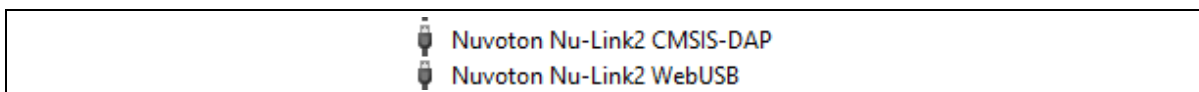
Query

4 CONFIGURE THE DEVICE

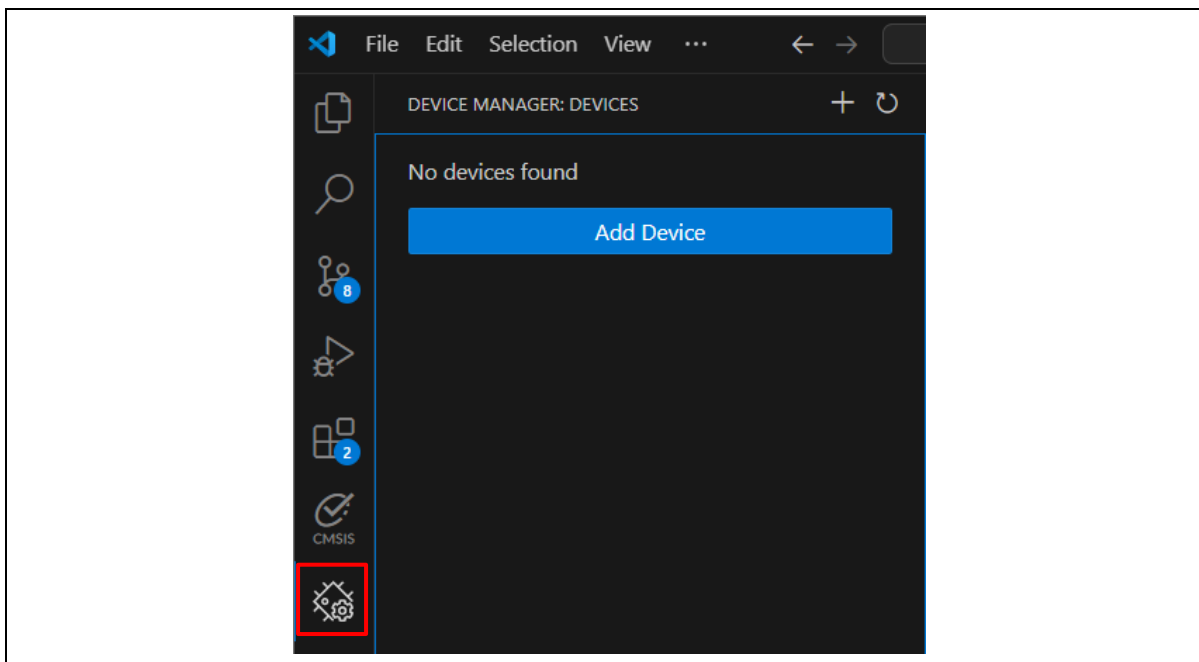
1. Install Nuvoton Nu-Link Keil Driver.
2. After installing the Keil driver, please check the box to install the Nu-Link USB Driver.



3. Installation complete in Windows Device Manager.



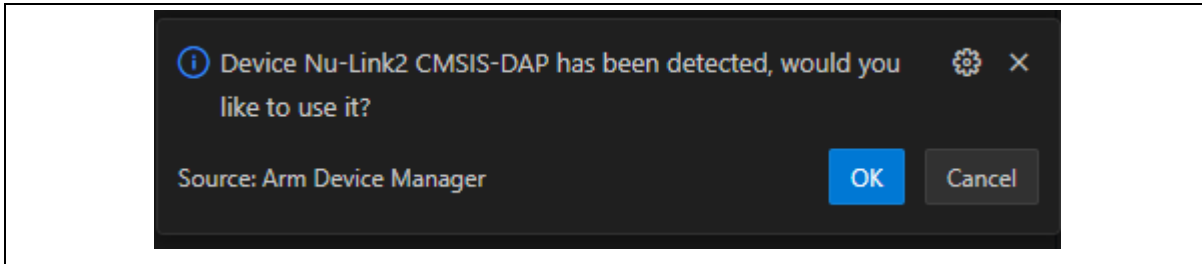
4. Click “**Device Manager**” icon in the Activity Bar to open the Device Manager.



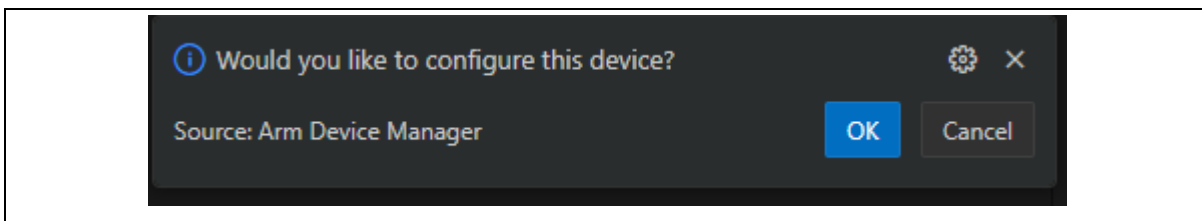
5. Connect device to your computer over USB.

The Device Manager detects the board and displays a pop-up message.

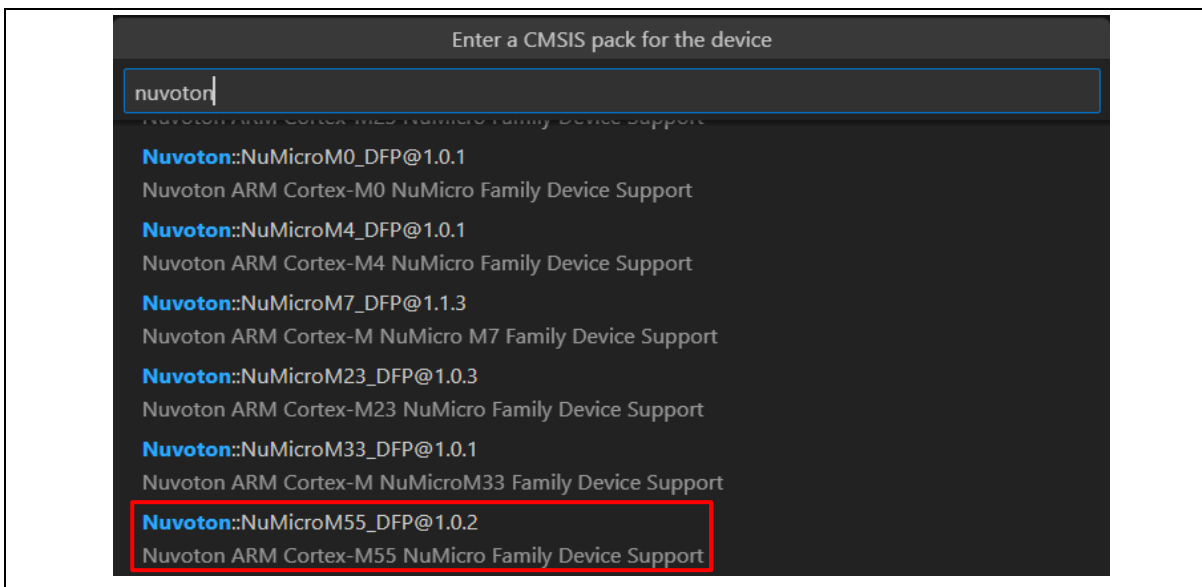
Press OK in the pop-up message and use it.



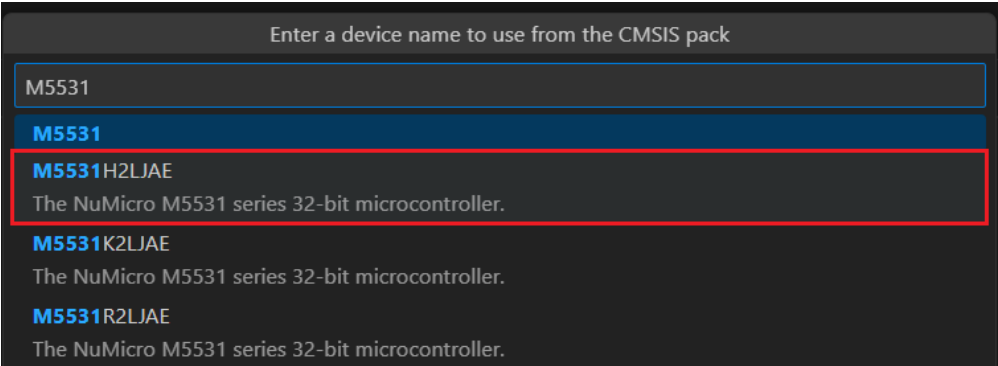
6. Press OK in the pop-up message and configure this device.



7. Text “**nuvoton**” in search bar and select corresponding CMSIS pack for the device.



8. Text your “**device model number**” in search bar.



Enter a device name to use from the CMSIS pack

M5531

M5531

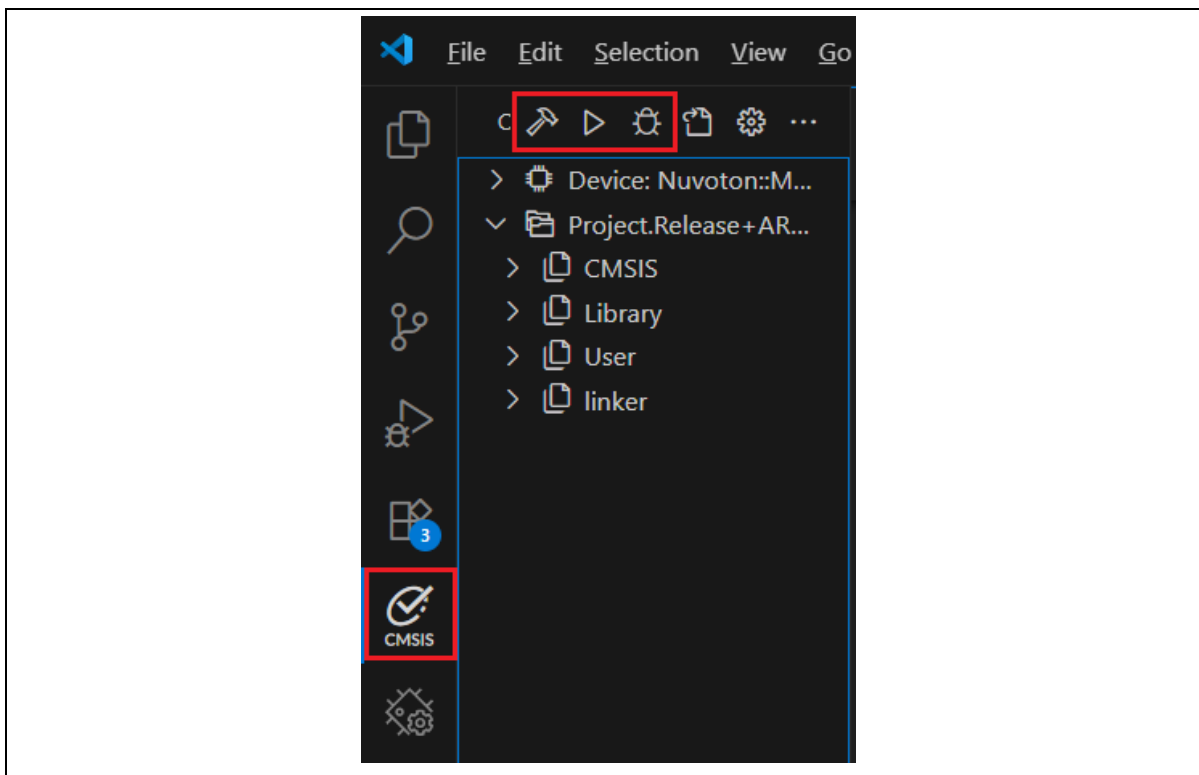
M5531H2LJAE
The NuMicro M5531 series 32-bit microcontroller.

M5531K2LJAE
The NuMicro M5531 series 32-bit microcontroller.

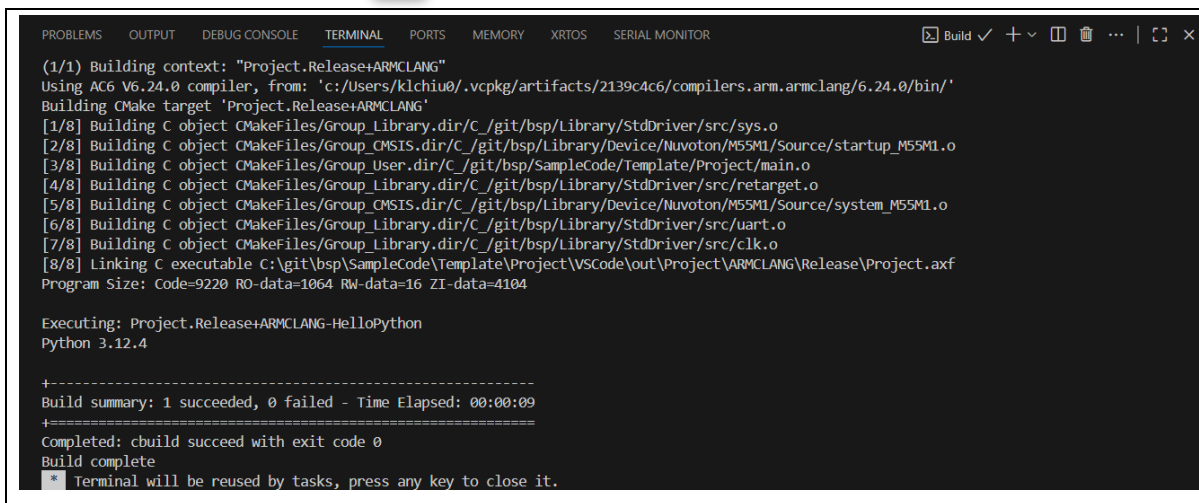
M5531R2LJAE
The NuMicro M5531 series 32-bit microcontroller.

5 RUN THE SAMPLE PROJECT

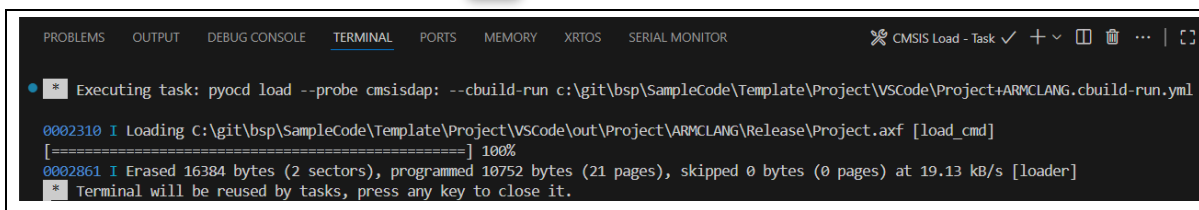
1. Click “CMSIS” icon in the Activity Bar.



2. Click “Build” button.

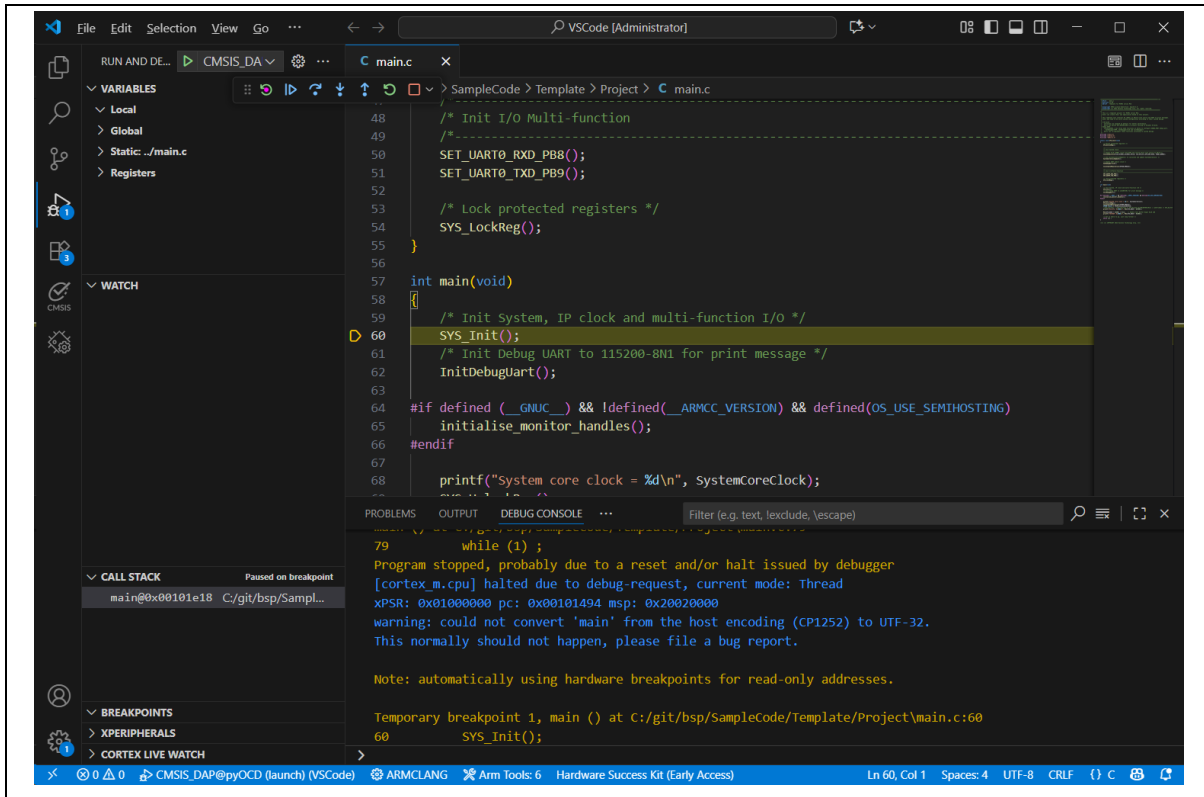


3. Click “Load & Run” button.

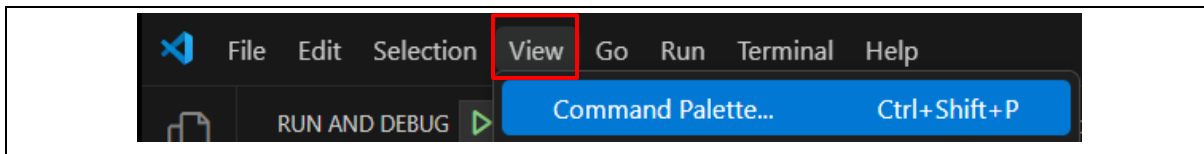


- Click “Load & Debug” button. 

The debugger stops at the main function.



- Click “View” menu and select “Command palette” item in the toolbar.



The screenshot displays the NuTool - PinView interface. The left pane shows the project structure with 'main.c' selected under 'C'. The middle pane contains the C source code for 'main.c', which includes system clock initialization, UART setup, and a loop that prints "Hello World". The right pane shows the pin connections for the code, listing various pins like UART_TX, UART_RX, and their corresponding peripheral functions.

REVISION HISTORY

Date	Revision	Description
2025.10.29	1.00	1. Initial version.

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